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Time-in-Range Analysis of Responses after Intravitreal Dexamethasone Therapy in Eyes with Diabetic Macular Edema

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Purpose: Time in range (TIR) is a novel end point that assesses the time during which an outcome remains within a predetermined range. Because the range includes normal parameters, it is indicative of clinically meaningful benefit. The MEAD trial comprised two 3-year randomized, multicenter, sham-controlled, phase III clinical studies that evaluated the efficacy and safety of dexamethasone intravitreal implant (DEX-I) in patients with diabetic macular edema. Dexamethasone intravitreal implant significantly improved best-corrected visual acuity (BCVA) and central retinal thickness (CRT) compared with sham treatment. We present a post hoc analysis of the MEAD trial to investigate TIR benefit across various thresholds of BCVA and CRT with DEX-I versus sham.

Design: Post hoc analysis of results from the randomized, multicenter, sham-controlled, phase III MEAD trial (NCT00168337 and NCT00168389).

Participants: Adults with type 1 or 2 diabetes mellitus and fovea-involved macular edema associated with diabetic retinopathy.

Intervention: Intravitreal injection of DEX-I 0.7 mg or sham procedure.

Main Outcome Measures: Time in range during year 1 was evaluated using BCVA thresholds of ≥69, ≥51, ≥59, and ≥64 letters, and CRT thresholds of <300, <353, <446, and <551 μm (the latter cutoffs being quartile [Q] 1, Q2, and Q3 of pooled baseline BCVA and CRT, respectively).

Results: Dexamethasone intravitreal implant 0.7 mg was associated with a statistically significant longer TIR versus sham at the BCVA ≥69-letter threshold (15.0 vs. 9.1 weeks, respectively; $P < 0.001$) and the CRT <300 μm threshold (18.5 vs. 8.3 weeks, respectively; $P < 0.001$). Dexamethasone intravitreal implant was also associated with longer TIR versus sham at thresholds of BCVA ≥59 (greater than or equal to Q2) and ≥64 letters (greater than or equal to Q3) and CRT <353 μm (less than Q1), <446 μm (less than Q2), and <551 μm (less than Q3) (all $P < 0.001$).

Conclusions: Patients receiving intravitreal DEX-I 0.7 mg had a longer time with BCVA above the driving threshold and below the normal limit of CRT during year 1 of the MEAD trial versus those who received sham. These results suggest that patients treated with dexamethasone experience a longer time with clinically meaningful outcomes than with sham, such as being able to drive or regaining normal structural retinal features.

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Diabetic retinopathy, including diabetic macular edema (DME), is a clinically important adverse outcome of diabetes mellitus (DM), which is associated with poor glycemic control. Diabetic macular edema is the leading cause of vision loss among patients with DM, with an estimated prevalence of 4.2% to 7.9% in patients with type 1 DM and 1.4% to 12.8% in patients with type 2 DM.[1] Importantly, the global prevalence of DM is projected to increase from 6.1% in 2021 to 9.8% in 2050,[2] which may lead to a corresponding increase in the size of the population affected by diabetic retinopathy and its complications.

Clinical trials of therapies for DME have assessed efficacy using various outcomes, including functional measures such as best-corrected visual acuity (BCVA) and structural measures such as central retinal thickness (CRT).[3–6] However, commonly used methods of assessment, such as change from baseline, proportion of patients with improvement, and area under the curve, do not fully capture outcomes.[7]

Time in range (TIR) is a novel end point that describes the proportion of time during which an outcome remains within a predetermined range. Typically, the selected range is indicative of clinically relevant benefit from treatment or

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meaningful benefit to patients' quality of life.[7,8] This approach provides additional context to the overall change in symptoms by measuring the time spent with benefit above a specified threshold (the proportion of time in the selected range out of the total time on trial is another meaningful parameter that accounts for potentially different durations of follow-up). It is particularly valuable where individual patients might have similar long-term changes in symptom ratings but have heterogeneous disease courses with marked short-term differences in magnitude and duration of improvement and worsening of symptoms relative to baseline.[7]

Intravitreally injected dexamethasone 0.7 mg implant (DEX-I; OZURDEX, DEX implant, Allergan, an AbbVie company) has been approved in multiple countries for the treatment of patients with DME.[9,10] The sustained-release formulation of the implant allows for prolonged retreatment intervals of ≥6 months,[3,11] reducing the burden of treatment on patients relative to the 1-month dosing intervals for the anti-VEGF therapies, ranibizumab and aflibercept.[4–6,12,13]

The approval of DEX-I was supported by the MEAD trial, which comprised two 3-year randomized, multicenter, sham-controlled, phase III clinical studies that evaluated the efficacy and safety of DEX-I in patients with DME.[3] Greater improvements from baseline in BCVA and greater reductions from baseline in CRT were seen in patients treated with DEX-I 0.7 mg and 0.35 mg, respectively, compared with sham treatment.[3]

Here, we present a post hoc analysis of the MEAD trial to evaluate TIR during the first year of treatment with DEX-I 0.7 mg versus sham across thresholds based on clinically meaningful visual acuity benefit (time with a BCVA above the US legal driving limit of ≥69 letters) or clinically meaningful reductions in CRT thickness (CRT <300 μm, the upper limit of normal, in the 1-mm central macular subfield of the study eye).[14,15] In addition, similar evaluations were carried out using quartiles (Qs) of baseline BCVA and CRT in the overall trial population as thresholds (BCVA ≥51, ≥59, and ≥64 letters and CRT <353, <446, and <551 μm) with the purpose of assessing the sensitivity of the observed differences between the groups.

Methods

Trial Design

The design of the MEAD trial has been described previously.[3] Briefly, the development program comprised 2 identically designed, randomized, multicenter, masked, sham-controlled phase III clinical studies (ClinicalTrials.gov NCT00168337 and NCT00168389) conducted between February 2005 and June 2012. Patients were randomized to treatment with DEX-I 0.7 mg, 0.35 mg, or a sham procedure in a 1:1:1 ratio; only patients randomized to DEX-I 0.7 mg or sham were included in this TIR analysis. Patient visits for both studies were scheduled at 1.5-month intervals during year 1 and at 3-month intervals during years 2 and 3 of treatment. Patients were eligible for retreatment with DEX-I if ≥6 months had elapsed since the most recent study treatment and there was evidence of residual retinal edema. Best-corrected visual acuity was evaluated using the ETDRS chart at every study visit. Central retinal thickness was derived from OCT scans conducted at 3-month intervals.[3]

The trial enrolled adult patients with type 1 or 2 DM and fovea-involved macular edema associated with diabetic retinopathy, which had been previously treated with medical or laser therapy. Treatment-naïve patients who refused laser treatment or would not benefit from laser treatment, in the opinion of the investigator, were also enrolled. Patients were required to have BCVA in the study eye of 34 to 68 letters (20/200–20/50) by the ETDRS method and CRT in the 1-mm central macular subfield of the study eye of ≥300 μm by OCT. Exclusion criteria included uncontrolled DM (glycosylated hemoglobin >10%); receipt of intravitreal anti-VEGF therapy within 3 months or intravitreal triamcinolone within 6 months of trial entry; and intraocular pressure (IOP) >23 mmHg without IOP-lowering medication, >21 mmHg with 1 IOP-lowering medication, or use of ≥2 IOP-lowering medications.[3]

The trial was conducted in accordance with the Declaration of Helsinki, and the trial protocol was approved by an institutional review board or independent ethics committee at each participating site. All patients provided written informed consent.

TIR Analysis

This post hoc analysis of the MEAD trial was performed to assess TIR for BCVA and CRT during year 1 in patients treated with DEX-I 0.7 mg and sham recipients. The TIR calculation has been described previously.[7] Briefly, TIR quantifies the absolute or relative amount of time that a patient's BCVA or CRT score remains above or below a clinically relevant threshold, respectively. Figure 1 depicts the concept of TIR using an idealized example of a potential subject. Patients randomized to DEX-I 0.35 mg, which has not been approved for use, were excluded. Time in range for BCVA was evaluated at a threshold of ≥69 ETDRS letters (the approximate legal limit for driving in the United States) and across thresholds based on quartile of pooled baseline BCVA (greater than or equal to Q1 = 51, greater than or equal to Q2 = 59, and greater than or equal to Q3 = 64 letters) for the DEX-I 0.7 mg and sham treatment groups. Time in range for CRT was evaluated at a threshold of <300 μm (normal limit for CRT) and across thresholds based on quartile of pooled baseline CRT (less than Q1 = 353 μm, less than Q2 = 446 μm, and less than Q3 = 551 μm) for both groups. The quartiles for BCVA score and CRT were based on baseline measurements and evaluated the

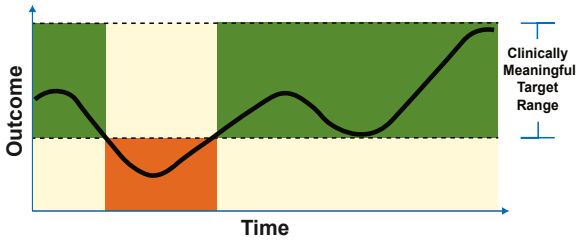


Figure 1. Time in range schematic illustration of an idealized example depicting the concept of TIR. Time is shown on the horizontal axis, and the outcome, such as BCVA score, is shown on the vertical axis. Time in range is the amount of time an outcome is above (green) or below (red) a predetermined threshold (in this case, above a predetermined threshold) that is considered to be clinically meaningful. The TIR is represented in green, while the out-of-range time is shown in red. BCVA = best-corrected visual acuity; TIR = time in range.

sensitivity of the TIR assessment in detecting differences between DEX-I and sham. Time in range was reported in weeks, and as the percentage of TIR over year 1 of the treatment period in the MEAD trial or actual annualized follow-up time (i.e., a TIR of 100% for BCVA means the patient spent the entire treatment period [1 year] at or above the threshold for BCVA).

Statistical Analyses

For the purposes of the TIR analysis, patients with an outcome assessment above the threshold for BCVA or below the threshold for CRT were considered to remain "in range" until the next assessment. Best-corrected visual acuity values above the threshold were counted as a 45-day time-in-effect range, and CRT values below the threshold were counted as a 90-day time-in-effect range based on the protocol-specified assessment intervals. Missing data were imputed using the last observation carried forward. Baseline BCVA- and OCT-adjusted analysis of covariance was used to evaluate the least square mean (LSM) differences in TIR outcomes between the DEX-I 0.7 mg and sham groups in the intention-to-treat (ITT). The model included baseline BCVA or CRT as covariates and treatment as a factor. A P value <0.05 was deemed statistically significant.

Results

Patients

A total of 1048 patients were randomized in the MEAD trial (ITT population) and received DEX-I 0.7 mg (n = 351), 0.35 mg (n = 347), or sham treatment (n = 350). All patients in the ITT population who received DEX-I 0.7 mg or sham treatment were included in the current post hoc analysis. Patient demographics and characteristics have been presented in detail previously[3] and were similar in the DEX-I 0.7 mg and sham groups, as shown in Table 1. Throughout the 1-year treatment period, all patients in the sham and DEX-I treatment arms had BCVA measurements taken at each trial visit. For CRT, a total of 97.7% of patients (n = 342/350) in the sham group, and 99.1% of patients (n = 348/351) in the DEX-I group had measurements taken at each trial visit (Table 2).

Table 1. Patient Demographics and Characteristics

Characteristic	DEX-I 0.7 mg (n = 351)	Sham (n = 350)
Age, yrs		
Mean [SD]	62.5 [8.3]	62.5 [9.5]
Range	33–85	26–88
Sex, n (%)		
Male	213 (60.7)	217 (62.0)
Female	138 (39.3)	133 (38.0)
Duration of diabetes, mean yrs [SD]	16.5 [9.0]	15.9 [9.1]
HbA1c, mean % [SD]	7.6 [1.2]	7.5 [1.1]
<8%, n (%)	233 (66.4)	249 (71.1)
>8%, n (%)	114 (32.5)	100 (28.6)
Not available	4 (1.1)	1 (0.3)
Duration of DME, mos		
Mean [SD]	23.6 [26.0]	25.9 [27.3]
Range	0–163	0–187
Baseline ETDRS letter score, mean [SD]	56.1 [9.9]	56.9 [8.7]
Baseline CRT, mean μm [SD]	463.0 [157.1]	460.9 [132.6]
Previous treatment for DME, n (%)		
Focal/grid laser	231 (65.8)	243 (69.4)
Intravitreal steroid	58 (16.5)	61 (17.4)
Anti-VEGF	25 (7.1)	26 (7.4)
None	104 (29.6)	89 (25.4)

CRT = central retinal thickness; DEX-I = dexamethasone intravitreal implant; DME = diabetic macular edema; HbA1c = proportion of glycated hemoglobin; SD = standard deviation.

TIR for BCVA

Time in range at the BCVA ≥69-letter threshold and across thresholds based on quartile of baseline BCVA in the pooled groups is shown in Figure 2, and treatment difference in Table 3. Time in range at the BCVA ≥69-letter threshold was longer in the DEX-I group (LSM 15 weeks; 95% confidence interval [CI], 13.52–16.47) than in the sham group (LSM 9.1 weeks; 95% CI, 7.65–10.61). The LSM difference between the DEX-I and sham groups was 5.9 weeks (95% CI, 3.78–7.95), which was statistically significant ($P < 0.001$). The LSM percentage of time with an in-range BCVA of ≥69 letters in the first year of treatment was 33.3% (95% CI, 30.05–36.61) in the DEX-I group, compared with 20.3% (95% CI, 17.01–23.58) in the sham group. The LSM difference between the DEX-I and sham groups was 13% (95% CI, 8.39–17.68), and this effect was statistically significant ($P < 0.001$). Dexamethasone intravitreal implant was associated with significantly longer TIR than sham at the BCVA ≥59 and ≥64 letter (greater than or equal to Q2 and greater than or equal to Q3) thresholds (both $P < 0.001$; Fig 2); however, no significant difference between DEX-I and sham was observed at the BCVA ≥51-letter (greater than or equal to Q1) threshold ($P = 0.092$). Therefore, the LSM treatment difference appeared to increase with increasing threshold, whereas TIR across thresholds increased as TIR cutoff decreased. Similarly, the frequency distributions (Fig S1A, available at www.ophthalmologyscience.org) and violin plots (Fig S1B) showing all the TIR points for BCVA at a cutoff of 69

Table 2. Number of Patients with BCVA and CRT Measurements at Each Trial Visit

Characteristic and Trial Visit	Sham (n = 350)	DEX-I 0.7 mg (n = 351)
BCVA, n (%)		
Mo 1.5	350 (100)	351 (100)
Mo 3	350 (100)	351 (100)
Mo 4.5	350 (100)	351 (100)
Mo 6	350 (100)	351 (100)
Mo 7.5	350 (100)	351 (100)
Mo 9	350 (100)	351 (100)
Mo 10.5	350 (100)	351 (100)
CRT, n (%)		
Mo 3	342 (97.7)	348 (99.1)
Mo 6	342 (97.7)	348 (99.1)
Mo 9	342 (97.7)	348 (99.1)

BCVA = best-corrected visual acuity; CRT = central retinal thickness; DEX-I = dexamethasone intravitreal implant.

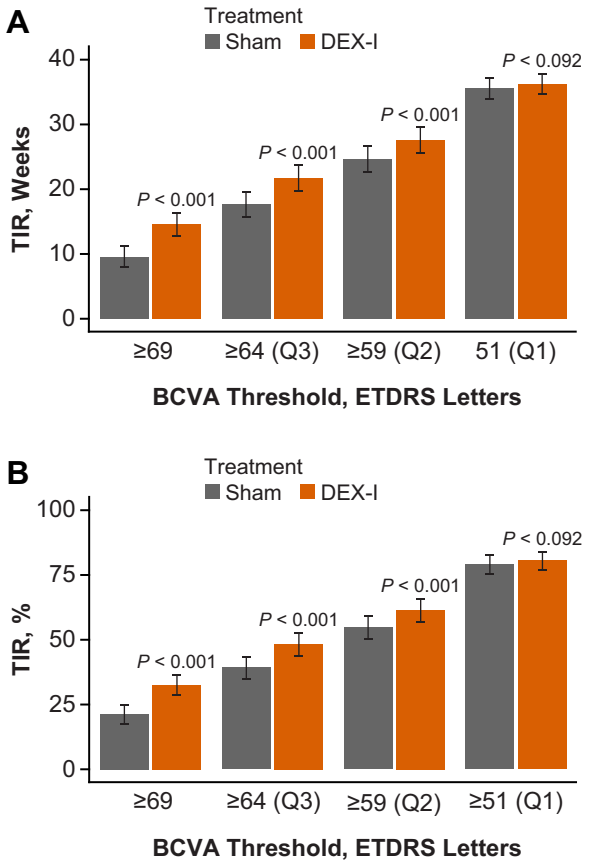


Figure 2. Time in range for BCVA with DEX-I (0.7 mg) vs. sham **(A)** in weeks and **(B)** as a percentage of year 1 at BCVA thresholds of ≥69 letters and quartiles of pooled baseline BCVA. Time in range for BCVA in weeks **(A)** or percentage of time during year 1 of treatment **(B)** are shown (y-axis) for the various BCVA thresholds (x-axis) for patients randomized to receive 0.7 mg DEX-I (orange) or sham (gray). Data represent the least square means, and error bars represent the 95% confidence intervals. Quartile 1, Q2, and Q3 thresholds were based on quartiles of pooled baseline BCVA levels. Thresholds for Q1, Q2, and Q3 were 51, 59, and 64 letters, respectively. BCVA = best-corrected visual acuity; DEX-I = dexamethasone intravitreal implant; Q = quartile; TIR = time in range.

letters differed between the sham and DEX-I treatment groups. Although the general distribution of patients was similar, most patients in the sham group spent no time during the trial period with a BCVA score ≥69 letters, with a median TIR of 0 weeks. In contrast, the frequency distribution for DEX-I was shifted toward a longer TIR, with a median TIR of 14.3 weeks. This suggests that, at the individual patient level, TIR for BCVA generally increased after the administration of DEX-I (please refer to the violin plots in Fig S1B for a visual representation of all the data points).

TIR for CRT

Time in range at the CRT <300 μm threshold and across thresholds based on quartile of baseline CRT is shown in Figure 3, and LSM treatment difference in Table 4. Time in range at the CRT <300 μm threshold (representing how long patients had a dry retina) was longer in the DEX-I group (LSM 18.5 weeks; 95% CI, 17.13−19.96) than in the sham group (LSM 8.3 weeks; 95% CI, 6.84−9.7). The LSM difference between the DEX-I and sham groups was 10.3 weeks, which was statistically significant (95% CI, 8.26−12.29; $P < 0.001$). The LSM percentage of time with an in-range CRT during the first year of treatment was 48.1% (95% CI, 44.41−51.76) in the DEX-I group and 21.4% (17.73−25.15) in the sham group. The LSM difference between the DEX-I and sham groups was statistically significant (26.6%; 95% CI, 21.43−31.87; $P < 0.001$). Evaluation of TIR across thresholds based on quartile of pooled baseline CRT showed a trend for increasing TIR with increasing threshold in both the DEX-I and sham groups. Dexamethasone intravitreal implant was associated with significantly longer TIR than sham across all thresholds (all $P < 0.001$; Fig 3). In addition, the TIR difference between the groups was greater at the lower CRT thresholds, with the maximal difference observed at 353 μm (Table 4). Frequency distributions (Fig S2A, available at www.ophthalmologyscience.org) and violin plots (Fig S2B) showing all the TIR points for CRT <300 μm differed between the sham and DEX-I treatment groups. The majority of patients in the sham group spent no time during the trial period with a CRT measurement <300 μm, with a median TIR of 0 weeks. In contrast, patients in the DEX-I group showed a clear increase in TIR for CRT at a cutoff of 300 μm, with a median TIR of 18.5 weeks. This suggests that, at the individual patient level, TIR for CRT generally increased following administration of DEX-I (please refer to the violin plots in Fig S2B for a visual representation of all the data points).

Discussion

The primary analysis of the MEAD trial at 3 years' follow-up showed that treatment with DEX-I 0.7 mg resulted in a ≥15-letter improvement in BCVA in a significantly greater proportion of patients compared with sham, as well as significant reductions in CRT relative to sham.[3] Based on those results, DEX-I was approved for the treatment of patients with DME.[16] The post hoc analysis of the MEAD trial presented here provides additional insight by showing that patients who received DEX-I 0.7 mg spent more time with BCVA above the legal limit for driving in the United States and CRT below normal thickness during the first year of treatment, indicating marked benefits for patients in parameters that affect their quality of life.

Importantly, the 2 most stringent thresholds of functional and structural benefit used in this analysis, BCVA ≥69 letters and CRT <300 μm, correspond to clinically meaningful functional benefit and normalization of structural retinal features.[15,17] Thus, the finding of significantly longer TIR with DEX-I compared with sham treatment translates to additional time with benefit that has a meaningful impact on patients' everyday lives, such as being able to continue driving during the first year of treatment. Furthermore, the consistent trend for greater TIR for DEX-I versus TIR for sham across additional thresholds based on patients'

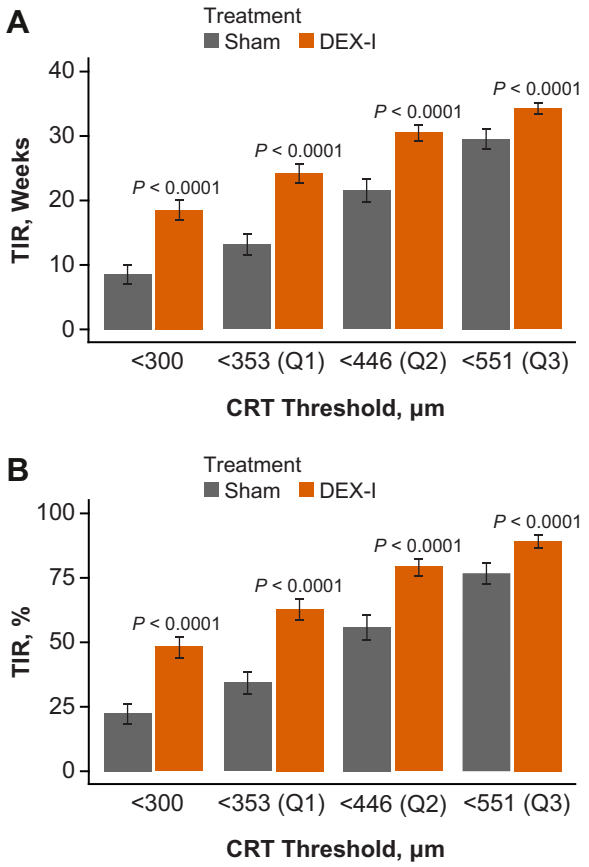


Figure 3. Time in range for CRT with DEX-I (0.7 mg) vs. sham **(A)** in weeks and **(B)** as a percentage of year 1 at CRT thresholds of <300 µm and quartiles of pooled baseline CRT. Time in range for CRT in weeks **(A)** or percentage of time during year 1 of treatment **(B)** are shown (y-axis) for the various CRT thresholds (x-axis) for patients randomized to receive 0.7 mg DEX-I (orange) or sham (gray). Data are represented as the least square means, and error bars represent the 95% confidence intervals. Quartile 1, Q2, and Q3 thresholds were based on quartiles of pooled baseline CRT levels. Thresholds for Q1, Q2, and Q3 were 353, 446, and 551 µm, respectively. CRT = central retinal thickness; DEX-I = dexamethasone intravitreal implant; Q = quartile; TIR = time in range.

symptom severity at baseline adds to the analysis by measuring time with benefit that, although of lower magnitude, is relevant to the population evaluated in the study. The magnitude of between-group differences in TIR also appeared to increase with increasing stringency of cutoffs, consistent with the observation of significantly greater improvement in BCVA and CRT in the DEX-I group versus the sham group in the primary analysis of the MEAD trial.[3]

As TIR is a novel end point in ophthalmology, only 1 trial utilizing this approach has been published to date. Proof of concept of the TIR end point was demonstrated in a post hoc analysis of the randomized Diabetic Retinopathy Clinical Research Retina Network Protocol T trial, which compared the VEGF inhibitors aflibercept, ranibizumab, and bevacizumab in patients with DME.[7] The analysis found that patients receiving aflibercept had significantly longer TIR than did patients receiving ranibizumab or bevacizumab at

Table 3. Time in Range LSM Difference for BCVA with DEX-I vs. Sham at a BCVA Threshold of ≥69 Letters and Thresholds Based on Quartiles of Pooled Baseline BCVA

	LSM Difference in TIR, DEX-I vs. Sham		
BCVA Threshold	*Wks (95% CI)*	*Percentage of Time (95% CI)*	*P* Value
≥69 letters	5.9 (3.78–7.95)	13 (8.39–17.68)	<0.001
≥51 letters (Q1)	1.5 (−0.25 to 3.33)	3.4 (−0.56 to 7.4)	0.092
≥59 letters (Q2)	4.2 (2.19–6.23)	9.4 (4.87–13.85)	<0.001
≥64 letters (Q3)	5.2 (3.09–7.4)	11.7 (6.86–16.45)	<0.001

BCVA = best corrected visual acuity; CI = confidence interval; DEX-I = dexamethasone intravitreal implant; LSM = least square mean; Q = quartile; TIR = time in range.
Q1, Q2, and Q3 thresholds were based on quartiles of pooled baseline BCVA.

the BCVA ≥69-letter threshold (LSM 41.2; 95% CI, 39.7–42.8), 37.6 (95% CI, 35.9–39.3), and 37.2 weeks (95% CI, 35.5–42.8) for aflibercept, ranibizumab, and bevacizumab, respectively; $P < 0.01$ for aflibercept versus both ranibizumab and bevacizumab.[7] However, these results cannot be directly compared with those of the analysis presented here due to inherent differences in trial design (primarily the time intervals of assessment) and patient populations that limit the resolution of the TIR assessments.

This analysis is subject to several limitations. First, although our post hoc analyses were performed using the overall ITT population, there remains a need to evaluate TIR as a prespecified end point in prospective clinical trials. Second, the frequency of BCVA and CRT assessments in the MEAD trial (1.5-month intervals for BCVA and 3-month intervals for CRT) limits the granularity with which TIR could be measured in this analysis. Because TIR was calculated using a last observation carried forward approach, patients who had an in-range BCVA or CRT assessment were assumed to have remained in range until the time of the subsequent assessment. This highlights the importance of frequent assessment as a design consideration for future studies to allow accurate assessment of TIR-based outcomes in patients whose symptom severity varies over short

Table 4. Time in Range LSM Difference for CRT with DEX-I vs. Sham at a CRT Threshold of <300 µm and across Thresholds Based on Quartiles of Pooled Baseline CRT

	LSM Difference in TIR, DEX-I vs. Sham		
CRT Threshold	*Wks (95% CI)*	*Percentage of Time (95% CI)*	*P* Value
<300 µm	10.3 (8.26–12.29)	26.6 (21.43–31.87)	<0.001
<353 µm (Q1)	11.4 (9.44–13.46)	29.7 (24.48–34.88)	<0.001
<446 µm (Q2)	9.2 (7.34–11.14)	24 (19.04–28.87)	<0.001
<551 µm (Q3)	5 (3.44–6.58)	13 (8.93–17.05)	<0.001

CI = confidence interval; CRT = central retinal thickness; DEX-I = dexamethasone intravitreal implant; LSM = least square mean; Q = quartile; TIR = time in range.
Q1, Q2, and Q3 thresholds were based on quartiles of pooled baseline CRT. Data were not available for 2 patients in the DEX-I group and 1 patient in the sham group.

timescales. Finally, the timeframe for this analysis was restricted to the first year of treatment because of the relatively higher rate of trial discontinuation from month 12 onward in the sham group compared with the DEX-I groups, precluding meaningful evaluation of TIR at later timepoints.

In conclusion, this post hoc analysis of the MEAD trial shows that patients receiving DEX-I 0.7 mg spent more time with functional and structural benefit during the first year of treatment than those who received sham. Importantly, patients who received DEX-I experienced longer time with in-range BCVA and CRT at thresholds linked to ability to drive and anatomical normalization compared with those who received no treatment. The contribution of TIR analysis in this and similar studies is that we learn the duration of time with benefit that has a meaningful impact on patients' everyday lives during the studied period.

Data Sharing

AbbVie is committed to responsible data sharing regarding the clinical trials we sponsor. This includes access to anonymized, individual, and trial-level data (analysis data sets), as well as other information (e.g., protocols, clinical trial reports, or analysis plans), as long as the trials are not part of an ongoing or planned regulatory submission. This includes requests for clinical trial data for unlicensed products and indications. These clinical trial data can be requested by any qualified researchers who engage in rigorous, independent, scientific research and will be provided following review and approval of a research proposal, statistical analysis plan, and execution of a data sharing agreement. Data requests can be submitted at any time after approval in the United States and Europe and after acceptance of this manuscript for publication. The data will be accessible for 12 months, with possible extensions considered. For more information on the process or to submit a request, visit the following link, then select "Home": https://vivli.org/ourmember/abbvie/

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Author Contributions:

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Data collection: Kozak, Do

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Overall responsibility: Kozak, Do, Lai, Ogidigben, López

Abbreviations and Acronyms:
BCVA = best-corrected visual acuity; **CI** = confidence interval; **CRT** = central retinal thickness; **DEX-I** = dexamethasone intravitreal implant; **DM** = diabetes mellitus; **DME** = diabetic macular edema; **IOP** = intraocular pressure; **ITT** = intention-to-treat; **LSM** = least square mean; **Q** = quartile; **TIR** = time in range.

Keywords:
Dexamethasone intravitreal implant (DEX-I), Diabetic macular edema (DME), Time in range.

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